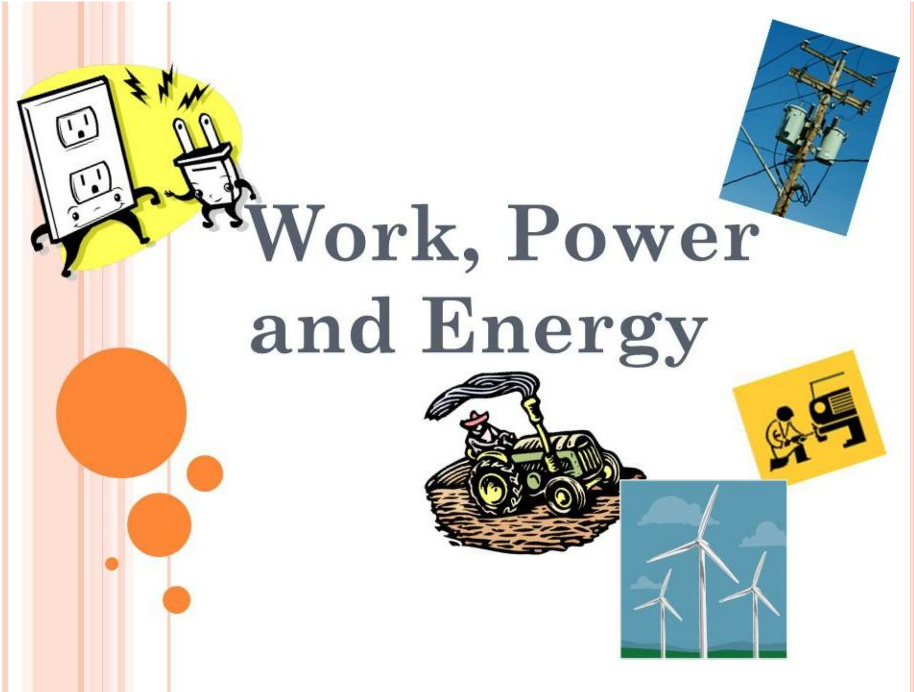
1 WORK, POWER AND ENERGY



LEARNING OUTCOMES

The learner should be able to:

- know that the sun is our major source of energy, and the different forms of energy.
- know that energy can be changed from one form into another and understand the law of conservation of energy.
- understand the positive and negative effects of solar energy.
- understand the difference between renewable and non-renewable energy resources with respect to Uganda.
- know and use the relationship between work done, force, and distance moved, and time taken.
- understand that an object may have energy due to its motion or its position and change between kinetic and positional potential energy.
- know the mathematical relationship between positional potential energy and kinetic energy, and use it in calculations.
- understand the meaning of machines and explain how simple machines simplify work.
- understand the principles behind the operation of simple machines.

INTRODUCTION

Work:

Work is done when you apply a force to move something over a distance. The S.I unit for work done is a joule(J).

$$workdone = force \times distance$$
$$W.D = f \times d$$

Example: Lifting a bag of groceries, pushing a car, or carrying a backpack upstairs all involve doing work.

Power

Power is how quickly you can do work or how fast you can use energy. The S.I unit for power is a watt(W).

$$power = \frac{workdone}{time}$$
$$P = \frac{W.D}{t} = \frac{f \times d}{t} = f \times V$$

Example: If you climb stairs quickly, you're using more power than if you climb slowly, even if you do the same amount of work.

Energy

Energy is the ability to do work or cause a change. It's what makes things happen.

Example: Your body has energy, and when you eat food, you're adding more energy. This energy is used when you run, play, or even think.

Forms of energy

Thermal (Heat) Energy:

Definition: The internal energy in a substance due to the motion of its atoms and molecules. Example: Boiling water, the warmth from a campfire, or the heat from a radiator.

Chemical Energy:

Definition: Energy stored in the bonds between atoms and molecules in chemical compounds. Example: Energy released during a chemical reaction, like burning wood or digesting food.

Electrical Energy:

Definition: The movement of electrons through a conductor. Example: Electricity flowing through wires, powering electronic devices.

Nuclear Energy:

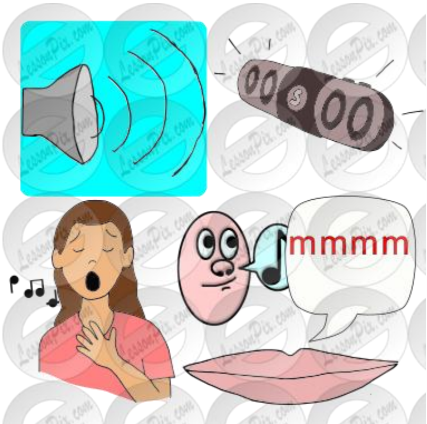
Definition: Energy stored in the nucleus of an atom and released during nuclear reactions. Example: Nuclear power plants use controlled nuclear reactions to generate electricity.

Light Energy:

Definition: Energy carried by electromagnetic waves, including light. Example: Sunlight,bulbs

Sound Energy:

Definition: Energy produced by the vibrations of particles in a medium (such as air, water, or solids). Example: The sound of a bell ringing, music playing, or someone speaking.



Mechanical energy:

This is the energy of motion.

(a) Kinetic energy

This is the energy possessed by a body due to its motion. Examples of such bodies include running water, moving bullet etc.

$$K.E = \frac{1}{2}MV^2$$

(b) Potential energy

This is the energy possessed by a body due to its position. For example a body raised above the ground has gravitational potential energy while a stretched spring has elastic potential energy.

$$P.E = mgh$$

ACTIVITIES

1. Calculate the kinetic energy of a 1000-kg car traveling at 17m/s?

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   ....

2. If a 0.2-kg ball is thrown with a velocity of 6m/s, what is its K.E?

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   ....

3. What is the potential energy of a 0.3-kg ball lifted to a height of 5.0 m above the ground?

   ....

   ....

4. A 1.5-kg book on top a shelf has a P.E of 29.4 .

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   ....

5. How high is the shelf ?

   ....

   ....

6. A machine lifts a loads of 2500N through a vertical height of 3m in 1.5s Find;

   (a) the power developed by a machine.

   ....

   ....

   (b) Using the same power how long would it take to lift of 6000N through a vertical height of 5m.

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7. A ball of 1 kg bounces off the ground to a height of 5m . find the energy lost

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8. A pump is rated 400W . How many kilograms of water can it raise in one hour through a height of 72m?

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9. A girl of mass 50kg climbs a flight of a stair having 40steps each of 30m high in 10 s. Calculate the power developed by the girl.

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10. Two men James and Peter were tasked to transfer bags of cement. Peter lifted 2 bags each weighing 50 kg and carried them through a distance of 4 metres. James carried one bag from the ground floor to the first floor using 150 stairs of height 0.02 metres each.

    By calculation, of the two men who did more work?

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    ....

11. A force of 600N pulls a load through distance of 20m in 2 minutes. Calculate the power expanded.

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    ....

12. A car moving at $30ms^{-1}$ pulls another with a force of 500N. Calculate the power of the pulling car.

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    ....

13. A 200g ball falls from a height of 0.2m. Calculate the kinetic energy just before it hits the ground.

    ....

    ....

14. A block of mass 2kg falls freely from rest through a distance of 3m. Find the kinetic energy of the block.

    ....

    ....

15. Cathy of mass 40kg runs up a stair case in 16 seconds. If each stair case is 20 cm high and she uses 100 Js^{-1} of her energy per second . Find the number of stairs.

    ....

    ....

16. A force of 50N moves an object through a distance of 200m in 40s. Find the work done.

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17. A stone of mass 20g falls through a distance of 10m. Calculate the kinetic energy it loses.

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18. When a force F acts on a body, the body's kinetic energy increases by 800 joules over a distance of 2metres.

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19. A car of mass 1.5×10^3kg climbs a hill in 900s. If the top if the hill is 50m above the starting point, find the average power output of the engine.

....

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20. A crane lifts 4 bricks per minute through a height of 1.5m. Find the power that is expanded if each brick weighs 100N.

....

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21. A man whose mass is 75kg climbs up a ladder of 6.5m high in 5s. Calculate:
    (a) work done
    (b) power expended

22. A block of mass 2kg falls freely from rest through a distance of 3m. find the kinetic energy of the block.

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23. An object of mass 2kg is dropped from the top of a building, hits the ground with kinetic energy of 900J. Calculate the height of the building.

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24. A bullet of mass 50g is fired at a speed of $400ms^{-1}$. How much energy does it have?

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25. The diagram in the figure shows large smooth bowl. Explain the energy changes that take place as the ball is released from P.

26. (a) Define the energy possessed by the bodies in A and B.

    (b) Name the two types of energy sources with two examples for each

27. A homeowner is concerned about reducing energy consumption. Explain how understanding the concepts of power and energy can help the homeowner make informed decisions about energy-efficient appliances and practices.

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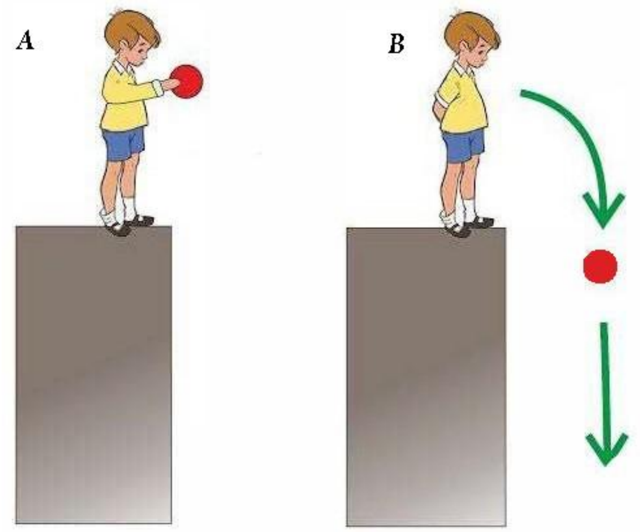

28. A town is considering adopting renewable energy sources for electricity generation. Discuss the advantages and disadvantages of different renewable energy options, such as solar, wind, and hydroelectric power, in terms of energy production and environmental impact.

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29. An automotive engineer is designing a more fuel-efficient car engine. Explain how concepts of work, power, and energy can be applied to optimize the engine's efficiency and reduce fuel consumption.

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    ....

30. A scientist is developing advanced solar cells for energy harvesting. Describe the energy changes occurring in them.

31. An engineer is tasked with designing a hydropower plant. Discuss the physics of how falling water is converted into electricity, including the role of turbines, generators, and potential energy.

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32. A roller coaster designer wants to create an exhilarating ride. Explain how the concepts of kinetic and potential energy are used in roller coaster design to create thrilling and safe experiences for riders.

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(b) You push a food tray 1.5 m along a cafeteria table with a constant force of 18 N. How much work do you do ?

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(c) An electric bulb is rated 100W. How much electrical energy does the bulb consume in 2 hours.

....

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33. **Complete each of the following statements or sentences**

(a) A gardener pushes on the angled handle of a lawn mower, causing it to move forward across a lawn. The only portion of the gardener's force that does work on the lawn mower is the force in the ....direction.

(b) A newton-meter is a measure of work also known as the ....

(c) The amount of work done in lifting a 25-N bag of sugar 2 meters is the same as lifting two 25-N bags of sugar..................... meter(s).

(d) The force applied to a machine is called the .... force.

(e) A machine makes work easier by multiplying force or.................., or by changing direction.

(f) All machines have a(n).................. of less than 100%.

(g) The ideal mechanical advantage would equal the actual mechanical advantage if there were no losses due to ....

(h) The efficiency of an actual machine is always less than..................... %.

(i) The output work of a certain machine is 12,600 J. If the input work is 18,000 J, the efficiency is ....

(j) When you use a paint can opener to open a can of paint, you use the paint can opener as a simple machine called a(n) ....

(k) A jar lid is an example of a simple machine called a(n) ....

(l) A screwdriver is a simple machine called a(n) ....

(m) A ramp in a parking garage is an example of a simple machine called a(n) ....

(n) The ideal mechanical advantage of a compound machine is the .... of the ideal mechanical advantages of the simple machines that make it up.

(o) Lengthening a ramp will .... its ideal mechanical advantage.

(p) A chef sometimes holds the tip of a knife stationary when chopping food. Held this way, the knife is a compound machine made up of a wedge and a ....

(q) As you wave your hand at the wrist, your hand is acting as a simple machine called a(n) ....

(r) As you bite into a peach, your front teeth act as a simple machine called a(n) ....

(s) Power is equal to.................. divided by time.

(t) A device that is twice as powerful as another can do........................the amount of work in the same amount of time.

34. **Modified True/False**
Indicate whether the sentence or statement is true or false. If false, change the identified word or phrase to make the sentence or statement true.

(a) Holding a 25N bag of sugar 1 meter above the floor requires 25 joules of work. ....

(b) The force exerted by a machine is called the output force. ....

(c) The mechanical advantage of a machine that changes only the direction of a force is 1. ....

(d) Efficiency compares the output work to the output force. ....

(e) A wheel and axle is a compound machine. ....

(f) A second-class lever always multiplies distance. ....

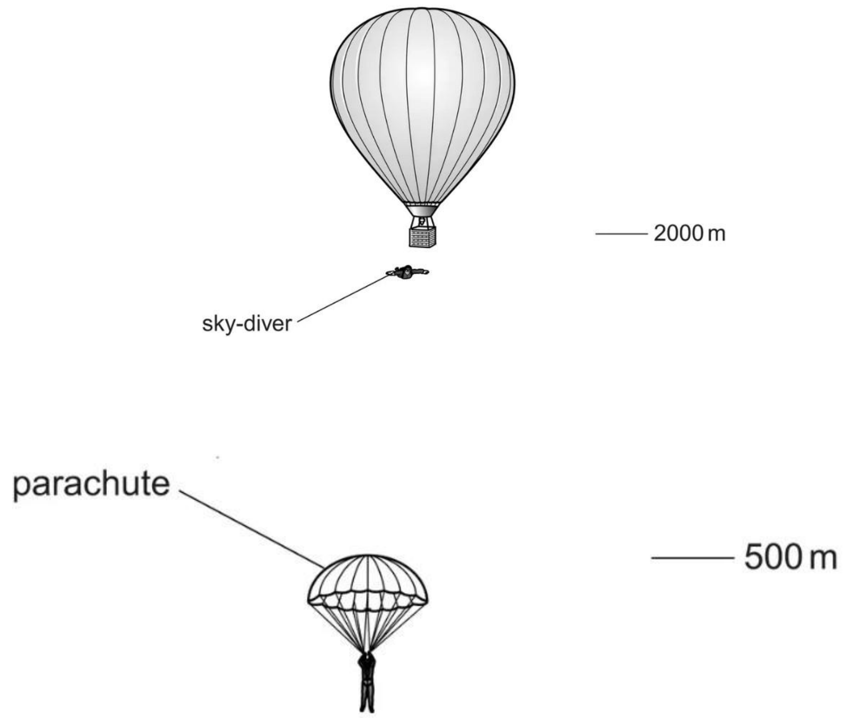
(g) The ideal mechanical advantage of a wheel and axle is the radius of the wheel times the radius of the axle. ....

(h) When you raise your leg, the knee acts as a fulcrum for the upper leg. ....

(i) You do work on an object when you lift it from the floor to a shelf. ....

(j) Energy is the rate at which work is done. ....

35. the figures below show the descent of a sky-diver from a stationary balloon.



The sky-diver steps from the balloon at a height of 2000 m and accelerates downwards. His speed is 52m/s at a height of 500m. He then opens his parachute. From 400 m to ground level, he falls at constant speed. The total mass of the sky-diver and his equipment is 92 kg.

Calculate, for the sky-diver,

(a) loss of gravitational potential energy in the fall from 2000 m to 500 m.

(b) The kinetic energy at a height of 500m

(c) The kinetic energy at 500 m is not equal to the loss of gravitational potential energy. Explain why there is a difference in the values.

(d) State what happens to the air resistance as the diver falls from 2000m to 500m.

(e) using a diagram show the forces acting on the diver.